Journal of Clinical Epidemiology 165 (2024) 111189

Journal of Clinical Epidemiology

GRADE GUIDANCE SERIES

GRADE guidance 37: rating imprecision in a body of evidence on test accuracy

Reem A. Mustafa[a,b,*], Ibrahim K. El Mikati[c], M. Hassan Murad[d], Monica Hultcrantz[e], Karen R. Steingart[f], Bada Yang[g,h], Mariska M.G. Leeflang[i], Elie A. Akl[b,j], Philipp Dahm[k], Holger J. Schünemann[b,l,m,n]

[a]*Division of Nephrology and Hypertension, Department of Internal Medicine, University of Kansas Medical Centre, 3901 Rainbow Blvd, MS3002, Kansas City, KS 61160, USA*

[b]*Department of Health Research Methods, Evidence, and Impact, McMaster University, 1280 Main Street West, Hamilton, Ontario L8S 4K1, Canada*

[c]*Outcomes and Implementation Research Unit, Department of Internal Medicine, University of Kansas Medical Center, Kansas City, KS, USA*

[d]*Division of Public Health, Infectious Diseases and Occupational Medicine, Mayo Clinic, 200 1st, ST, SW, Rochester, MN 55902, USA*

[e]*Department of Learning, Informatics, Management and Ethics, Karolinska Institutet, Tomtebodav. 18 A, SE-171 77 Stockholm, Sweden*

[f]*Department of Clinical Sciences, Liverpool School of Tropical Medicine, Pembroke Place, Liverpool L3 5QA, UK*

[g]*Julius Center for Health Sciences and Primary Care, University Medical Center Utrecht, Utrecht University, Utrecht, The Netherlands*

[h]*Department of Epidemiology and Data Science, Amsterdam University Medical Centers, University of Amsterdam, Meibergdreef 9, 1105, AZ, Amsterdam, The Netherlands*

[i]*Clinical Epidemiology and Biostatistics and Bioinformatics Academic Medical Center, University of Amsterdam, Meibergdreef 9, P.O.Box 227001100 DE, Amsterdam, The Netherlands*

[j]*Department of Internal Medicine, American University of Beirut, P.O. Box 11-0236, Beirut, Lebanon*

[k]*Minneapolis VA Health Care System, Urology Section 112D, One Veterans Drive, Minneapolis, MN 55417, USA*

[l]*Department of Medicine, McMaster University, 1280 Main Street West, Hamilton, L8S 4L8 Ontario, Canada*

[m]*Department of Biomedical Sciences, Humanitas University, Via Rita Levi Montalcini 4, 20090 Milano, Italy*

[n]*Institute for Evidence in Medicine, Medical Center and Faculty of Medicine, University of Freiburg, Breisgau, Germany*

Accepted 13 October 2023; Published online 28 October 2023

Abstract

Objectives: To provide guidance on rating imprecision in a body of evidence assessing the accuracy of a single test. This guide will clarify when Grading of Recommendations Assessment, Development and Evaluation (GRADE) users should consider rating down the certainty of evidence by one or more levels for imprecision in test accuracy.

Study Design and Setting: A project group within the GRADE working group conducted iterative discussions and presentations at GRADE working group meetings to produce this guidance.

Results: Before rating the certainty of evidence, GRADE users should define the target of their certainty rating. GRADE recommends setting judgment thresholds defining what they consider a very accurate, accurate, inaccurate, and very inaccurate test. These thresholds should be set after considering consequences of testing and effects on people-important outcomes. GRADE's primary criterion for judging imprecision in test accuracy evidence is considering confidence intervals (i.e., CI approach) of absolute test accuracy results (true and false, positive, and negative results in a cohort of people). Based on the CI approach, when a CI appreciably crosses the predefined judgment threshold(s), one should consider rating down certainty of evidence by one or more levels, depending on the number of thresholds crossed. When the CI does not cross judgment threshold(s), GRADE suggests considering the sample size for an adequately powered test accuracy review (optimal or review information size [optimal information size (OIS)/review information size (RIS)]) in rating imprecision. If the combined sample size of the included studies in the review is smaller than the required OIS/RIS, one should consider rating down by one or more levels for imprecision.

* Corresponding author. University of Kansas Medical Center, 3901 Rainbow Blvd, MS3002, Kansas City, KS 66160, USA. Tel.: +1-913-588-6048; fax: +1-913-588-6074.

E-mail address: rmustafa@kumc.edu (R.A. Mustafa).

https://doi.org/10.1016/j.jclinepi.2023.10.005

0895-4356/© 2023 Published by Elsevier Inc.

Conclusion: This paper extends previous GRADE guidance for rating imprecision in single test accuracy systematic reviews and guidelines, with a focus on the circumstances in which one should consider rating down one or more levels for imprecision. © 2023 Published by Elsevier Inc.

Keywords: GRADE; Imprecision; Threshold; Systematic review; Guideline; Test accuracy

1. Introduction

GRADE introduced guidance for rating the certainty in test accuracy and testing interventions in 2008 and updated its guidance in 2020 [1−4]. These articles, however, did not provide a detailed operationalization for rating the domain of imprecision related to test accuracy. For this domain, GRADE did provide guidance only for interventions [5]. That guidance introduced the use of the optimal information size (OIS) (also called review information size [RIS]) to assess imprecision [5,6].

Updated GRADE guidance for rating imprecision related to interventions introduced rating down for imprecision by one, two, or three levels (serious, very serious, and extremely serious) [7,8]. Methods for rating imprecision in the field of network meta-analysis have evolved as well [9]. Another development was the recognition that rating certainty of evidence requires defining a target of certainty rating, which determines how imprecision and other GRADE domains are rated [6,10,11].

The thinking around GRADE in the context of tests and diagnosis has advanced with the emergence of new publications. In 2017, members of the GRADE diagnosis and tests project group published a series of papers detailing methods, challenges, and perspectives around health care-related tests and diagnostic test strategies (hereafter to be called tests) [12−16]. Then the GRADE working group published the evidence to decision (EtD) framework for decisions about tests [17], operationalizing the certainty judgment for test accuracy studies [2,3], and introducing ranges for certainty ratings [11] for tests. Additionally, multiple groups have applied GRADE and published guidelines about tests. Examples include the American Society of Hematology (ASH) guidelines on the diagnosis of Von Willebrand disease (VWD) [18], ASH guidelines on the diagnosis of venous thromboembolism [19], the World Health Organization (WHO) guidelines on the screening for cervical intraepithelial metaplasia, WHO guideline on the diagnosis of tuberculosis [20], and the Infectious Diseases Society of America (IDSA) COVID-19 diagnostic guidelines for molecular, serology, and antigen testing [21−24].

This GRADE guidance paper provides detailed guidance on rating imprecision in a body of evidence assessing test accuracy with a focus on the accuracy of a single test. This guidance applies to scenarios when the available evidence is limited to test accuracy studies and there is no direct evidence assessing the effect of a test on people-important outcomes [1]. This guide should be applied when systematic reviewers, health technology assessors, guideline developers, and other decision makers are assessing the certainty in test accuracy evidence using judgment thresholds about the level of accuracy of the test (e.g., very accurate [VA], accurate [A], inaccurate [I], and very inaccurate [VI]), although we also refer to situations in which systematic review authors may not want to set more than one threshold.

This guide does not apply to evidence about comparisons of test accuracy [25], and it also does not apply when guideline developers perform formal modeling and produce effect estimates on people-important outcomes while considering test results as variables in a model. In the latter case, certainty ratings can be based on performing sensitivity analyses by calculating the effect estimate using the extremes of a test accuracy confidence interval (CI).

2. Methods

This project was conducted under the auspices of the GRADE diagnosis and tests project group. To produce this article, we followed the methods for developing official GRADE articles that we published recently [26]. This began with determining that this guidance was a priority within the GRADE test and diagnosis project group which draws its members from the bigger GRADE working group. As all GRADE project groups, the group has specific terms of reference and goals that are regularly reviewed. A smaller group of the authors produced initial clarity about the concepts that underwent multiple rounds of review in small and larger group meetings with members of the GRADE working group before a draft article was produced. The initial guide included guiding principles and a five-step process (see Appendix 1), with the intention of acknowledging areas that require further empiric evidence in the future.

The initial guidance underwent several rounds of feedback and revisions from experts in systematic review methodology, biostatisticians, and systematic reviewers, both with and without experience in test accuracy reviews. These revisions took place through online teleconferences, small group sessions during GRADE working group meetings, workshops and focus groups, a large-group session at the GRADE working group meeting held in Hamilton, Canada, in June 2019, and during the US GRADE network biannual workshop in Colorado, USA in March 2019. Lead authors considered the feedback and, after each round,

What is new?

Key findings

- When rating the certainty of evidence for a body of evidence assessing test accuracy of a single test, Grading of Recommendations Assessment, Development and Evaluation (GRADE) recommends explicitly defining the target of certainty rating by prespecifying one or more judgment thresholds. Using the confidence interval (CI) approach, one should consider rating down for imprecision by one or more levels depending on the number of judgment threshold(s) crossed (e.g., crossing three thresholds implies rating down imprecision by three levels).
- In scenarios where the CI of test accuracy estimates does not cross predefined judgment threshold(s), GRADE users are advised to consider the sample size for an adequately powered test accuracy review (optimal or review information size (OIS/RIS) in rating imprecision.
- When inconsistency exists, OIS/RIS calculation may guide in deciding when it is appropriate to only rate down for inconsistency vs. rate down for both inconsistency and imprecision.

What this adds to what was known?

- This paper provides GRADE guidance on rating imprecision in a body of evidence assessing test accuracy, providing specific guidance on circumstances when, using one, two, or three or three judgment thresholds, one should consider rating down the certainty of evidence for imprecision by one, two, or three levels by considering the CI approach and the required sample size for an adequately powered test accuracy review (OIS/RIS).

What is the implication and what should change now?

- When rating the certainty of evidence for test accuracy, GRADE users should consider rating down for imprecision by one or more levels depending on the number of predefined judgment thresholds crossed by the CIs for absolute test accuracy results (true and false positives and negatives) and, when no threshold is crossed, the OIS/RIS.
- Additional work is needed for guidance related to rating imprecision in comparative test accuracy evidence.

made necessary changes. We made additional revisions in keeping with most up-to-date GRADE imprecision guidance for interventions [7,8]. We then applied the approach in several test accuracy reviews and guidelines to test feasibility and detect potential challenges [18,19,21,22,27].

We presented the final draft framework to the GRADE working group at the meeting held in Krakow, Poland for discussion in July 2022 and a revised version during the virtual GRADE working group meeting in November 2022. During that meeting, the GRADE working group approved the guidance before it was approved by the GRADE guidance group.

3. Results

3.1. Defining concepts

To draw conclusions or make decisions about tests, paired test accuracy statistics such as sensitivity and specificity are usually applied to a hypothetical cohort with a specified prevalence of the target condition to calculate absolute test accuracy results: true positive (TP), false negative (FN), true negative (TN), and false positive (FP). Box 1 includes definitions of key terms used in this paper. This calculation can be applied when sensitivity and specificity are consistent across different populations and there are no concerns for spectrum bias.

We use the term *judgment threshold* to refer to the acceptable number or proportion of true and false negative and positive results to support conclusions for systematic reviews and health care decisions for guidelines. In scenarios where systematic reviews or health technology assessments choose to rely on a single judgment threshold in order to determine whether or not to evaluate this test, that threshold could be between A and I. On the other hand, when decision makers decide to establish multiple judgment thresholds, it would correspond to thresholds between VA and A, between A and I, and between I and VI. However, the choice and number of the thresholds should be described by the rater considering the relevant context. This is not to be confused with the positivity threshold (i.e., cut-off value) at which a test result is deemed positive. Judgment thresholds should be defined after considering the consequences (subsequent management and testing) based on test results and their effect on people-important outcomes. There may be one or more judgment thresholds, depending on the setting. In section 4.3 below, we elaborate on how GRADE users can define judgment thresholds.

3.2. The importance of contextualization when rating imprecision in test accuracy

GRADE introduced the concept of contextualizing the ranges of certainty for both test and treatment questions

Box 1 Definitions of key terms used in this paper

- Sensitivity is the probability of a positive test given that the person has the condition.
- Specificity is the probability of a negative test given that the person does not have the condition.
- TP, TN, FN, and FP rates in a simulated cohort: proportion of people with TP, TN, FN, or FP results in a hypothetical cohort. These rates are calculated by applying the paired accuracy statistics (e.g., sensitivity and specificity) to a hypothetical cohort with specific prevalence and of a particular size (e.g., 1,000 people).
- Judgment threshold: maximal acceptable rate of true and FN and FP results that influence conclusions drawn from the body of evidence by systematic reviewers or decision makers.
- Optimal information size (also called review information size [OIS/RIS]): 'The number of patients generated by a conventional sample size calculation for a single adequately powered study using a plausible effect size or (in the case of the RIS) a threshold and acceptable type 1 and 2 error rates. This number is compared to the total number of patients included in a meta-analysis across all studies [6,28]. OIS/RIS is also the required sample size for an adequately powered test accuracy review.

[10,11,29]. The *minimally contextualized* approach considers the line of no effect or the minimally important difference to guide certainty judgment for each outcome separately. The *fully contextualized* approach accounts for all critical people-important outcomes, bearing in mind the decision(s) that need to be made and the associated values [11]. When addressing the question of imprecision in test accuracy studies, we suggest that contextualized decisions are required. Instead of discussing concepts such as minimally or partially contextualized approaches [11,29], the focus of this guidance is on establishing judgment threshold(s) (See Box 2). This guidance also highlights that even if the CI of sensitivity or specificity is narrow, downgrading may still be required if no judgment can be made about whether the test is sufficiently A based on the judgment threshold(s).

3.3. Establishing judgment threshold(s) of interest for test accuracy systematic reviews and guidelines

Before rating the certainty, authors need to establish the judgment threshold(s). Note that in other GRADE articles the term *decision threshold* is used [8,31]. This is also the first decision point for judging imprecision in test accuracy studies (see decision 1 of Fig. 1A and B). Initial

Box 2 Rationale for using judgment threshold(s) instead of discussing the level of contextualization

When addressing the question of imprecision in test accuracy studies, we suggest that contextualized decisions are required. Instead of discussing concepts such as minimally or partially contextualized approaches, the focus of this guidance is on establishing judgment threshold(s) for the following reasons:

First, in test accuracy research, there is no "line of no effect". One may think that the line of no effect for test accuracy is sensitivity or a specificity of $\leq$50%: in other words, when the test performs as or worse than a coin toss. However, the tradeoff between sensitivity and specificity and the fact that tests may be used in different circumstances highlight that certain tests may be useful even if their sensitivity or specificity is below 50% [20]. We propose that 50% should not be used as a replacement for the line of no effect in decisions based on test accuracy studies without considering the context for decision making. Additionally, using 'line of noninformative test' as the threshold may not be informative. For most tests that are being evaluated by systematic reviewers or guideline developers, it is likely that they are being considered as an option in the management of people with a condition.

Second, people-important outcomes are a consequence of a sequence of events including performing the test of interest, followed by a management strategy that may include other tests, treatment, or no follow-up, which in turn affects people-important outcomes. Take, for example, the case where people with an FN result (people mislabeled as not having the condition) will likely not receive treatment. In this scenario, the natural progression of the condition or disease might be critical (e.g., progression and metastasis of an undetected malignant tumor) or noncritical (e.g., benign tumor with no effect on function or quality of life). Establishing an analytical framework would allow for representing visually all those possibilities [1] and would be critical in guiding judgments about imprecision and certainty.

Third, there is an inherent multidirectional relationship between test accuracy results and people-important outcomes. Different people-important outcomes may be influenced by the same test accuracy results and multiple test accuracy results may affect one people-important outcome. For example, a person classified as FN may suffer all stages or consequences of a disease and the one classified as FP may suffer different adverse effects. With the existing trade-off between sensitivity and specificity, FN and FP results are typically not independent from each

other [30]. Therefore, setting even one threshold (e.g., maximal acceptable FN threshold) has implications on multiple people-important outcomes, both directly (i.e., consequences of lack of detection of disease when setting FN threshold) or indirectly (i.e., consequences of false detection and management of disease when setting FN threshold). At the same time, the same people-important outcome may be influenced by multiple test accuracy results. For example, mortality can be affected by FN results (e.g., metastasis of an undetected malignant tumor), FP results (e.g., complications of treating a disease-free individual), or TP results (e.g., death from malignancy despite treatment).

determination of the judgment threshold should not be influenced by the knowledge of the test accuracy estimates and their CIs (or credible intervals) to avoid biasing the establishment of thresholds.

Setting the judgment threshold(s) of interest is based on formal/informal modeling for critical people-important outcomes, which should consider the linked evidence (including evidence about natural disease progression). This approach focuses on the proportions of TP, TN, FN, and FP in a hypothetical cohort as well as the proportion of FN (FN divided by all those who have a negative test) and FP (FP divided by all those who have a positive test). The use of thresholds based on sensitivity or specificity may be problematic, as these measures do not take prevalence into account. Judgments about imprecision may differ across settings with varying prevalence. The same sensitivity and specificity will lead to different numbers of FPs and TPs and FNs when applied to different prevalence and hence different imprecision judgments.

Developing an analytic framework would allow visually representing all those possibilities [1]. Examples of analytic frameworks and decision trees include the WHO guidelines on cervical cancer screening and diagnosis [32], the US Preventive Task Force [33], ASH guidelines on the diagnosis of VWD [18], IDSA COVID-19 diagnostic guidelines for molecular tests [23]. We applied the GRADE for test accuracy approach and setting decision thresholds from a guideline panel on cows' milk allergy (CMA) in 2011 [34]. In that approach, the guideline panel was presented with case scenarios that described people-important outcomes and asked about when they would conduct testing and start treatment. We back calculated the thresholds for TP, TN, FN, and FP to guide the recommendations about testing for CMA. This approach would typically start with people-important outcomes and model the upstream management and testing strategies, finally reaching a modeled judgment threshold [35]. Performing this modeling in a formal fashion requires expertise and may be difficult to implement. Korevaar et al. [36] suggested a framework

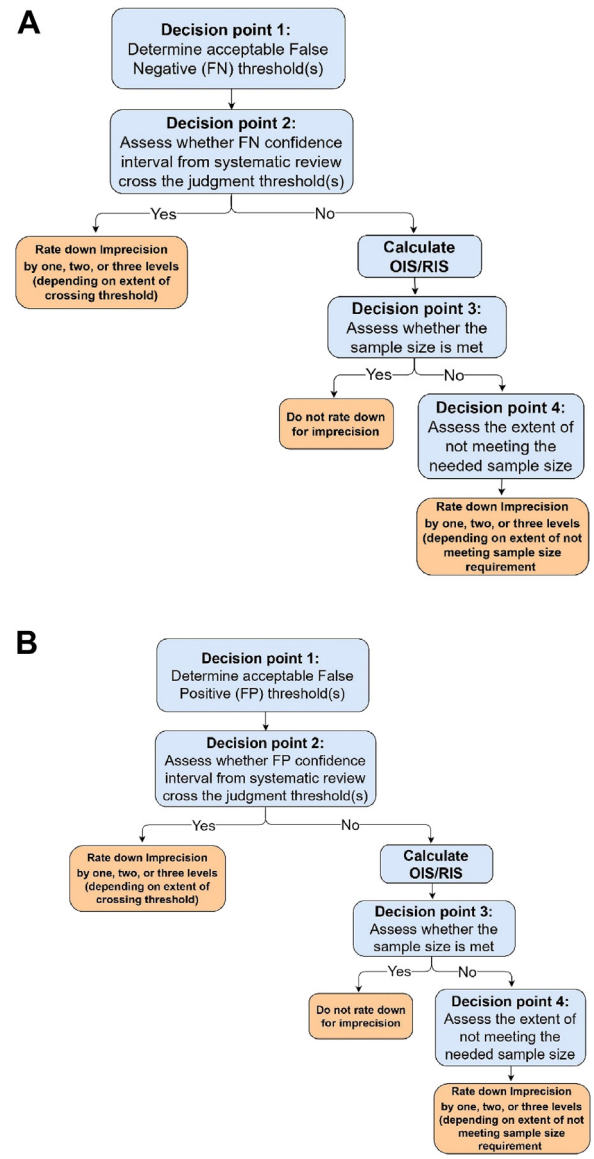

Fig. 1. (A) Algorithm summarizing the GRADE approach to rating imprecision in evidence assessing test accuracy (sensitivity) (B) Algorithm summarizing the GRADE approach to rating imprecision in evidence assessing test accuracy (specificity). Note: The first decision point is not specifically for the GRADE domain of imprecision, but for defining the target of the certainty rating that is relevant for all GRADE certainty domains. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

for defining minimally acceptable criteria for test performance that could also be applied when using informal modeling to establish a judgment threshold, for example by asking "Considering x number of people suspected of the target condition, how many FN and FP results are acceptable, considering the potential harms of such misclassifications?"

The evidence that links test accuracy results with people-important outcomes will likely affect the establishment of judgment threshold(s). Take, for example, the case

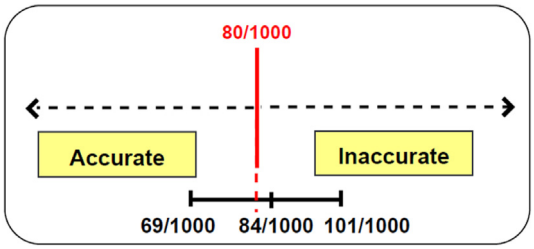

Fig. 2. Ventilation perfusion diagnostic test for pulmonary embolism—CI for proportion of FN (69–101 per 1,000) crossing a single threshold (80 per 1,000). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

of COVID-19 testing. Critical people-important outcomes may include hospitalization, complications, and death. In early stages of the COVID-19 pandemic (when there were higher hospitalization rates per infection, worse disease severity, no effective treatments, and no vaccine available) one would accept more stringent judgment thresholds for using a test as compared to later in the pandemic (when some treatments proved efficacious and when the disease severity lessened with vaccines and some newer variants). People-important outcomes remained the same, but due to the change in the linked evidence, the judgment threshold(s) became less stringent at later stages of the pandemic; decision makers were willing to accept higher judgment thresholds for FP and FN, especially for tests that are likely to be followed by additional confirmatory testing.

Based on the GRADE EtD framework, the judgments thresholds for test accuracy include: (1) threshold between VA and A; (2) threshold between A and I; and (3) threshold between I and VI.

3.4. One or multiple judgment thresholds

As stated previously, establishing judgment thresholds of interest is based on formal or informal modeling for critical people-important outcomes, often including natural progression as linked evidence. For guidelines, decision makers would typically use judgments thresholds to separate more than two levels of accuracy (e.g., VI, I, A, VA). The use of three previously mentioned judgment thresholds allows for aligning the imprecision judgment with the EtD framework. However, for systematic reviews and health technology assessments, GRADE users might prefer to explicitly establish only one judgment threshold (between A and I). In the setting of using a single judgment threshold, GRADE users must proceed with caution, considering potential implications on the judgment of imprecision, as described below.

Explicitly defining a single judgment threshold implies implicit definition of additional thresholds. Consider the case in which the maximal acceptable FN proportion is 5% (50 out of a hypothetical cohort of 1,000 people) and the FNs from the systematic review ranged between 350 and 450 per 1,000. In this situation, GRADE users might consider the accuracy estimate as I and precise or I and imprecize in the absence of a predefined judgment threshold. Rating as either precise or imprecize test accuracy must then rely on an implicitly defined threshold (I/VI).

3.5. The challenge of small studies with high accuracy

The first GRADE imprecision guidance recommended using OIS/RIS to assess imprecision in cases of large relative effects with modest sample size and number of events [5]. GRADE users have faced scenarios where the test of interest reported high accuracy when it was assessed only in a low number of participants (e.g., 50 participants). In these scenarios, the CI may not cross any of the established thresholds. This may give the impression of preciseness in the results when in reality larger studies may show less optimistic test accuracy results. To address the limitations of relying on CIs alone in such scenarios, we additionally suggest the use of the required sample size for an adequately powered diagnostic accuracy review (OIS/RIS) to inform judgments of imprecision. The degree to which CIs can be considered 'vulnerable' in diagnostic accuracy research is still a matter on ongoing discussion within the GRADE working group. We include OIS/RIS as part of this guide, consistent with the GRADE imprecision guidance for treatment. The OIS/RIS is a feasible approach for many users even if they have limited statistical knowledge.

The calculation of the OIS/RIS would require the knowledge of (1) prevalence of the target condition; (2) sensitivity/specificity from the systematic review; and (3) acceptable margin of error. The margin of error can be an

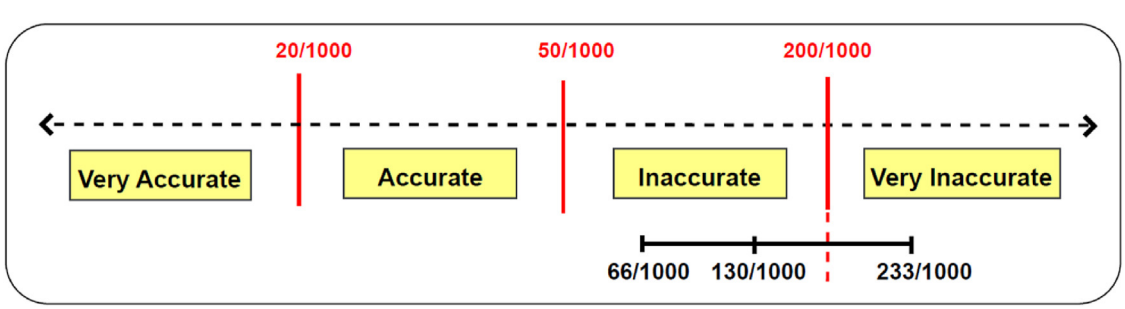

Fig. 3. Von Willebrand factor collagen binding (VWF: CB) diagnostic test for VWD 2A & 2B—CI for proportion of FN crossing a single threshold. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

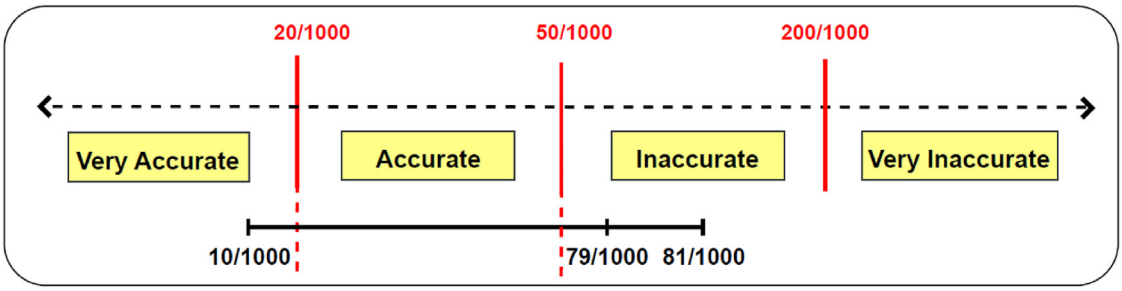

Fig. 4. Von Willebrand factor multimer analysis diagnostic test for VWD 2A & 2B—CI for proportion of FN crossing two thresholds. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

arbitrary difference (e.g., 0.10 difference in sensitivity). However, such difference would be better contextualized if it was back calculated from FPs and FNs proportions that have practical consequences to stakeholders. Guideline developers can develop consensus on FN and FP values that would be maximally acceptable or define a test that is accurate, VA, I, or VI (see the subsequent section about judgment thresholds). The margin of error can be derived from the difference between these judgment thresholds so that OIS/RIS calculation informs whether the analysis is sufficiently powered to discriminate between such differences. In appendix 2, we provide a link to an online calculator that facilitates this calculation as well as the rationale used for it.

3.6. Rating down imprecision by one, two, or three levels

Once judgments thresholds are defined, one should assess whether the CI of FN or FP crosses one or more thresholds (see decision point 2 of Fig. 1A and B). If the CI crosses judgment threshold(s), then rating down will depend on the number of crossed thresholds (i.e., rating down by one level for one crossed threshold, two levels for two crossed thresholds, and three levels for three crossed thresholds).

If only a single judgment threshold is explicitly defined and the CI crosses the threshold, then one should rate down for imprecision by at least one level. GRADE users can consider rating down by more than one level if the CI crosses additional implicitly defined thresholds.

3.6.1. Example 1: crossing one judgment threshold when one is explicitly defined

Consider a systematic review of diagnostic accuracy of ventilation perfusion scan for the diagnosis of pulmonary embolism [19] (see Fig. 2). After considering the consequences of the test results, the panel decided the judgment threshold (A/I) to be 80 FN per 1,000 tested individuals. A meta-analysis of 13 studies with 3,994 patients yielded a summary sensitivity of 58% and a summary FN proportion of 84 per 1,000 (95% CI: 69−101 per 1,000).

Because the FN CI crosses the threshold of 80 per 1,000 and thus the test may be either A or I, authors would rate down for imprecision. As mentioned previously, GRADE users can consider rating down by more than one level if the CI crosses additional implicitly defined thresholds.

3.6.2. Example 2: crossing one judgment threshold when three are explicitly defined

Consider a systematic review of diagnostic accuracy of VWD factor collagen binding for the diagnosis of VWD type 2A and 2B in patients suspected to have VWD type 2 [18] (see Fig. 3). After considering the consequences of the test results, the panel decided on the following FN judgment thresholds: VA/A at 20 per 1,000, A/I at 50 per 1,000, and I/VI at 200 per 1,000. A meta-analysis of nine studies with 476 patients yielded a summary sensitivity of 90% and

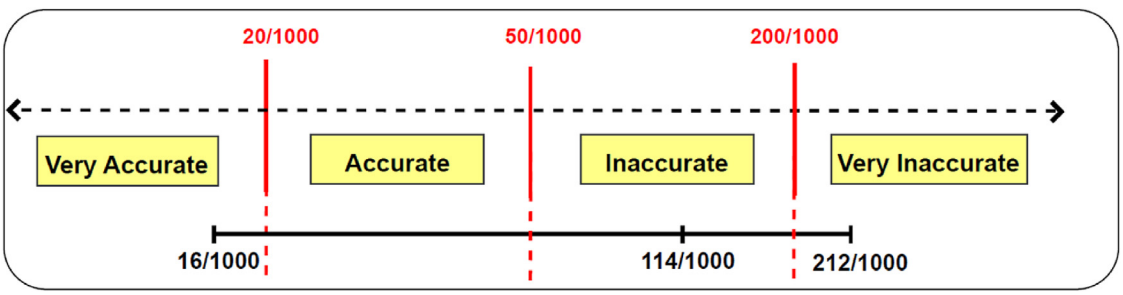

Fig. 5. Von Willebrand factor multimer analysis diagnostic test for VWD 2M—CI for proportion of FN crossing three thresholds. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

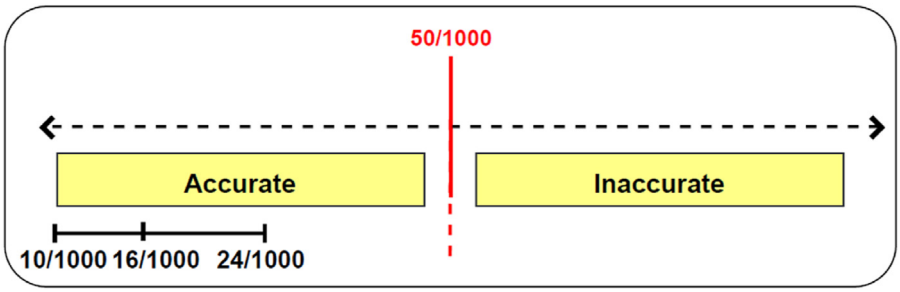

Fig. 6. Telehealth assessment diagnostic test for mild cognitive impairment in persons with suspected cognitive impairment in the secondary care setting—CI for proportion of FP in the A range. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

a summary FN proportion of 130 (95% CI: 66−233) per 1,000.

Because the CI of FN crosses the judgment threshold of 200 per 1,000, the test may be considered to be either I or VI. Thus, authors would rate down certainty by one level for imprecision.

3.6.3. Example 3: crossing two judgment thresholds when three are explicitly defined

Consider a systematic review of diagnostic accuracy of Von Willebrand factor multimer analysis for the diagnosis of VWD type 2A and 2B in patients suspected to have VWD type 2 [18] (see Fig. 4). After considering the consequences of the test results, the panel decided on the following judgment threshold for FN proportion: VA/A at 20 per 1,000, A/I at 50 per 1,000, and I/VI at 200 per 1,000. A meta-analysis of four studies with 283 patients yielded a summary sensitivity of 90% and a summary FN proportion of 79 (95% CI: 10−81) per 1,000.

Because the CI for FN crosses two judgment thresholds of 20 per 1,000 and 50 per 1,000 (i.e., it is not certain whether the diagnostic test is VA, A or I), authors would rate down certainty by two levels for imprecision.

3.6.4. Example 4: crossing three judgment thresholds when three are explicitly defined

Consider a systematic review of diagnostic accuracy of Von Willebrand factor multimer analysis for the diagnosis of VWD type 2M in patients suspected to have VWD type 2 [18] (see Fig. 5). After considering the consequences of the test results, the panel decided on the following judgment thresholds for FN proportion: VA/A at 20 per 1,000, A/I at 50 per 1,000, and I/VI at 200 per 1,000. A meta-analysis of four studies with 103 patients yielded a summary) sensitivity of 86% and summary FN proportion of 114 (95% CI: 16−212) per 1,000.

Because the CI for FN crosses three thresholds of 20 per 1,000, 50 per 1,000, and 200 per 1,000 (i.e., the diagnostic test is imprecisely VA, A, I, or VI), authors would rate down certainty by three levels for imprecision.

3.6.5. Example 5: calculating OIS/RIS when judgment threshold(s) are not crossed

Consider a systematic review of diagnostic accuracy of telehealth assessment for diagnosis of mild cognitive impairment in persons with suspected cognitive impairment in the secondary care setting [27] (see Fig. 6). After considering the consequences of the test results, the panel decided the judgment threshold for FP proportion (A/I) to be 50 per 1,000. A study with 100 patients yielded a specificity of 73% and FP proportion of 16 (95% CI: 10−24) per 1,000.

Because the CI for FP does not cross any threshold, GRADE users would consider performing the OIS calculation. In this case, considering a prevalence of 40%, specificity of 73%, and a judgment threshold of 50 per 1,000, the required sample size would be 727. This means that the sample size for OIS/RIS is not met. GRADE users may rate down certainty by more than one level depending on the degree to which the OIS requirement is not met. In this case, rating down by two levels may be appropriate. Additional examples for calculating OIS/RIS can be found under appendix 3.

3.7. Relationship between judging inconsistency and imprecision

In random effects models, which are commonly used in diagnostic test meta-analyses, the CI of the summary estimate is derived from both within and between study heterogeneity. Hence, when between-study heterogeneity is large, the CI of the summary estimate can be wide. This may lead to rating down certainty twice, once for inconsistency and once for imprecision, despite the correlation between these two domains. This ‘double counting’ may not be appropriate. Table 1 summarizes possible scenarios for rating inconsistency in relation to imprecision. Additional work is needed to further address the relationship between inconsistency and imprecision, and we hope this would be addressed in future methodological research that involves relevant stakeholders.

When review authors face important imprecision (CI of the summary estimate crossing one or more judgment thresholds), they need to first determine if there is an

Table 1. Possible scenarios for rating inconsistency in relation to imprecision

Scenarios	Overall summary (pooled) confidence interval?	Inconsistency?	Overall sample size from all included studies	Decision
Scenario A	Narrow	No	Large	Do not rate down for inconsistency or imprecision
Scenario B	Narrow	No	Limited	Consider rating down for imprecision only
Scenario C	Wide	No	Limited	Consider rating down for imprecision only
Scenario D	Wide	Yes	Large	Consider rating down for inconsistency only
Scenario E	Wide	Yes	Limited	Consider rating down for both inconsistency and imprecision

important inconsistency that warrants downgrading. This judgment about importance can be facilitated using the judgments thresholds and should not depend on statistical measures of heterogeneity such as I^2, which are usually elevated in diagnostic test meta-analyses and may lead to misleading conclusions. After a decision is made about rating down the certainty of evidence due to inconsistency, we recommend subsequently evaluating the OIS/RIS. If the available sample size across studies is lower than OIS/RIS, then it would be appropriate to lower certainty for both domains. If the sample size exceeds OIS/RIS and CIs of FP/FN proportions do not overlap test accuracy thresholds, then there is no need to lower certainty for imprecision. If the sample size exceeds OIS/RIS, this may be a situation when we opt to not rate down for both domains and only rate down for inconsistency. In this case, we suggest that raters provide an explanation that, though the presence of some imprecision was recognized, imprecision was attributed to inconsistency. Lastly, when CIs of summary estimates cross multiple judgment thresholds, imprecision is likely severe, and it would be appropriate to rate down for both domains.

4. Discussion

This GRADE guidance provides stepwise guidance for the rating of imprecision in test accuracy systematic reviews and guidelines. We show that defining judgment threshold(s) for the certainty ratings is crucial because the ranges chosen will affect the interpretation of the result and the degree of certainty. The establishment of any judgment threshold will require consideration of consequences of test results including further testing and effects of treatment on people-important outcomes. This will inevitably include assumptions, as the evidence that links test accuracy to people-important outcomes is typically scarce and unavailable [1]. For test accuracy evidence, one may end up judging that the test results are A and precise, A but imprecize, I, and precise, or I and imprecize. Certainty ratings do not imply the test should or should not be used. Additional factors should be taken into consideration before recommendations and decisions are made about the test. Decisions can, for example, depend on the relationship between sensitivity and specificity, what the alternative to the test strategy would be, or other factors from the EtD framework such as resource use.

4.1. Implications for practice

This paper suggests GRADE users incorporate OIS/RIS calculation in specific scenarios to address the need for caution when reaching high certainty based on studies that only included few people. Although there are no empirical data that supports the use of OIS/RIS in rating imprecision in test accuracy evidence, the GRADE working group's collective experience with test accuracy studies has identified cases of high accuracy based on studies with small number of patients. There are known challenges common to sample size calculations in test accuracy studies. For example, the sample size calculations used in this guide are based on asymptotic theory of normal approximation, which may not be appropriate when sensitivity and specificity are expected to be very high. In this case, the exact method would be more appropriate [37]. There are also various ways to determine the margin of error used for sample size calculations. The margin of error in this guide is informed by the judgment threshold instead of being arbitrary assigned. Although the margin of error may be informed by what studies suggest, it should not be solely data-driven, as that would be inconsistent with the OIS/RIS concept. Our approach also assumes that sensitivity and specificity are constant across different variance values, an assumption that may not be true in all settings [38]. A solution in this case would be to jointly model the disease prevalence along with sensitivity and specificity, as in the case in using trivariate generalized linear mixed models [39]. Further research is needed to explore all these alternative methodologies.

This guide aligns with recent guidance from the Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy [40]. The Handbook provides guidance for rating imprecision considering the number of included studies, number of participants informing the estimation of sensitivity, number of participants informing the estimation of specificity, and the statistical uncertainty around the

summary estimates (e.g., 95% CI), while acknowledging that the judgment of imprecision is subjective. GRADE, however, proposes using the OIS/RIS approach, a method for determining the combined sample size of studies when no judgment threshold is crossed.

4.2. Implications for research

This paper highlights multiple areas for future work. Additional work and empirical evidence is required for evaluating different methods of calculating OIS/RIS and how common scenarios are in which neither judgment thresholds are crossed and nor the OIS/RIS requirement is met. Work with authors of systematic reviews and decision-makers is also needed to address the relationship of the domains inconsistency and imprecision. Since the focus of this work is on the accuracy of a single test, more research is needed for guidance related to comparative test accuracy studies [41].

5. Conclusion

Addressing imprecision in a body of evidence on test accuracy requires considerations of the judgment threshold(s), how the CI of the summary estimate relates to those thresholds, and potentially the sample size for an adequately powered diagnostic accuracy study or review using a method such as OIS. We provided an approach for rating imprecision in a body of evidence about the accuracy of a single test. Additional work is needed for guidance related to comparative test accuracy studies [41].

CRediT authorship contribution statement

RAM, IKE, and HJS conceptualized this work. IKE collected examples from guidelines and systematic reviews. IKE and RAM drafted the initial manuscript. RAM, MHM, MH, KRS, BY, M.M.G.L, and HJS revised the manuscript. All authors provided input that resulted in important modifications. All authors approved the final manuscript.

Declaration of competing interest

The authors declare no direct financial conflict of interest. They are members or contributors to the GRADE working group. Multiple coauthors have served as methodologists on a number of systematic reviews or guidelines that provided examples for this manuscript.

Data availability

Examples came from published systematic reviews and guidelines.

Acknowledgments

We would like to thank GRADE Working Group members who gave valuable comments and suggestions on the draft of this article, especially Ingrid Arevalo Rodriguez. The authors also thank attendants of GRADE working group meetings who have contributed to the article during group discussions.

Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2023.10.005.

References

[1] Schunemann HJ, Mustafa RA, Brozek J, Santesso N, Bossuyt PM, Steingart KR, et al. GRADE guidelines: 22. The GRADE approach for tests and strategies-from test accuracy to patient-important outcomes and recommendations. J Clin Epidemiol 2019;111:69−82.
[2] Schünemann HJ, Mustafa RA, Brozek J, Steingart KR, Leeflang M, Murad MH, et al. GRADE guidelines: 21 part 1. Study design, risk of bias, and indirectness in rating the certainty across a body of evidence for test accuracy. J Clin Epidemiol 2020;122:129−41.
[3] Schünemann HJ, Mustafa RA, Brozek J, Steingart KR, Leeflang M, Murad MH, et al. GRADE guidelines: 21 part 2. Test accuracy: inconsistency, imprecision, publication bias, and other domains for rating the certainty of evidence and presenting it in evidence profiles and summary of findings tables. J Clin Epidemiol 2020;122:142−52.
[4] Schunemann HJ, Oxman AD, Brozek J, Glasziou P, Jaeschke R, Vist GE, et al. Grading quality of evidence and strength of recommendations for diagnostic tests and strategies. BMJ 2008;336:1106−10.
[5] Guyatt GH, Oxman AD, Kunz R, Brozek J, Alonso-Coello P, Rind D, et al. GRADE guidelines 6. Rating the quality of evidence—imprecision. J Clin Epidemiol 2011;64:1283−93.
[6] Schünemann HJ. Interpreting GRADE's levels of certainty or quality of the evidence: GRADE for statisticians, considering review information size or less emphasis on imprecision? J Clin Epidemiol 2016;75:6−15.
[7] Zeng L, Brignardello-Petersen R, Hultcrantz M, Mustafa RA, Murad MH, Iorio A, et al. GRADE Guidelines 34: updated GRADE guidance for imprecision rating using a minimally contextualized approach. J Clin Epidemiol 2022;150:216−24.
[8] Schünemann HJ, Neumann I, Hultcrantz M, Brignardello-Petersen R, Zeng L, Murad MH, et al. GRADE Guidance article 35: update on rating imprecision for assessing contextualized certainty of evidence and making decisions. J Clin Epidemiol 2022;150:225−42.
[9] Brignardello-Petersen R, Guyatt GH, Mustafa RA, Chu DK, Hultcrantz M, Schünemann HJ, et al. GRADE guidelines 33: addressing imprecision in a network meta-analysis. J Clin Epidemiol 2021;139:49−56.
[10] Hultcrantz M, Rind D, Akl EA, Treweek S, Mustafa RA, Iorio A, et al. The GRADE Working Group clarifies the construct of certainty of evidence. J Clin Epidemiol 2017;87:4−13.
[11] Hultcrantz M, Mustafa RA, Leeflang MMG, Lavergne V, Estrada-Orozco K, Ansari MT, et al. Defining ranges for certainty ratings of diagnostic accuracy: a GRADE concept paper. J Clin Epidemiol 2020;117:138−48.
[12] Schünemann HJ, Mustafa RA. Decision making about healthcare-related tests and diagnostic test strategies. Paper 1: a new series on testing to improve people's health. J Clin Epidemiol 2017;92:16−7.

[13] Mustafa RA, Wiercioch W, Cheung A, Prediger B, Brozek J, Bossuyt P, et al. Decision making about healthcare-related tests and diagnostic test strategies. Paper 2: a review of methodological and practical challenges. J Clin Epidemiol 2017;92:18−28.
[14] Mustafa RA, Wiercioch W, Falavigna M, Zhang Y, Ivanova L, Arevalo-Rodriguez I, et al. Decision making about healthcare-related tests and diagnostic test strategies. Paper 3: a systematic review shows limitations in most tools designed to assess quality and develop recommendations. J Clin Epidemiol 2017;92:29−37.
[15] Mustafa RA, Wiercioch W, Arevalo-Rodriguez I, Cheung A, Prediger B, Ivanova L, et al. Decision making about healthcare-related tests and diagnostic test strategies. Paper 4: international guidelines show variability in their approaches. J Clin Epidemiol 2017;92:38−46.
[16] Mustafa RA, Wiercioch W, Ventresca M, Brozek J, Schünemann HJ, Bell H, et al. Decision making about healthcare-related tests and diagnostic test strategies. Paper 5: a qualitative study with experts suggests that test accuracy data alone is rarely sufficient for decision making. J Clin Epidemiol 2017;92:47−57.
[17] Schünemann HJ, Mustafa R, Brozek J, Santesso N, Alonso-Coello P, Guyatt G, et al. GRADE Guidelines: 16. GRADE evidence to decision frameworks for tests in clinical practice and public health. J Clin Epidemiol 2016;76:89−98.
[18] James PD, Connell NT, Ameer B, Di Paola J, Eikenboom J, Giraud N, et al. ASH ISTH NHF WFH 2021 guidelines on the diagnosis of von Willebrand disease. Blood Adv 2021;5:280−300.
[19] Lim W, Le Gal G, Bates SM, Righini M, Haramati LB, Lang E, et al. American Society of Hematology 2018 guidelines for management of venous thromboembolism: diagnosis of venous thromboembolism. Blood Adv 2018;2:3226−56.
[20] World Health Organization. WHO consolidated guidelines on tuberculosis: module 3: diagnosis: rapid diagnostics for tuberculosis detection. Licence: CC BY-NC-SA 3.0 IGO. Geneva: World Health Organization; 2021.
[21] Hanson KE, Altayar O, Caliendo AM, Arias CA, Englund JA, Hayden MK, et al. The infectious diseases society of America guidelines on the diagnosis of COVID-19: antigen testing. Clin Infect Dis 2021:ciab557.
[22] Hanson KE, Caliendo AM, Arias CA, Englund JA, Hayden MK, Lee MJ, et al. Infectious diseases society of America guidelines on the diagnosis of COVID-19:serologic testing. Clin Infect Dis 2020: ciaa1343.
[23] Hanson KE, Caliendo AM, Arias CA, Hayden MK, Englund JA, Lee MJ, et al. The infectious diseases society of America guidelines on the diagnosis of COVID-19: molecular diagnostic testing. Clin Infect Dis 2021:ciab048.
[24] Hanson KE, Caliendo AM, Arias CA, Englund JA, Lee MJ, Loeb M, et al. Infectious diseases society of America guidelines on the diagnosis of COVID-19. Clin Infect Dis 2020:ciaa760.
[25] Yang B, Mustafa RA, Bossuyt PM, Brozek J, Hultcrantz M, Leeflang MMG, et al. GRADE Guidance: 31. Assessing the certainty across a body of evidence for comparative test accuracy. J Clin Epidemiol 2021;136:146−56.
[26] Schünemann HJ, Brennan S, Akl EA, Hultcrantz M, Alonso-Coello P, Xia J, et al. The development methods of official GRADE articles and requirements for claiming the use of GRADE - a statement by the GRADE guidance group. J Clin Epidemiol 2023;159:79−84.
[27] McCleery J, Laverty J, Quinn TJ. Diagnostic test accuracy of telehealth assessment for dementia and mild cognitive impairment. Cochrane Database Syst Rev 2021;7:Cd013786.
[28] Pogue J, Yusuf S. Overcoming the limitations of current meta-analysis of randomised controlled trials. Lancet 1998;351:47−52.
[29] Zeng L, Brignardello-Petersen R, Hultcrantz M, Siemieniuk RAC, Santesso N, Traversy G, et al. GRADE guidelines 32: GRADE offers guidance on choosing targets of GRADE certainty of evidence ratings. J Clin Epidemiol 2021;137:163−75.
[30] Bossuyt PM, Deeks JJ, Leeflang MM, Takwoingi Y, Flemyng E. Preface. In: Deeks JJ, Bossuyt PM, Leeflang MM, Takwoingi Y, editors. Cochrane handbook for systematic reviews of diagnostic test accuracy version 2Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy. Version 2.0 (updated July 2023). London: Cochrane; 2022. Available at https://training.cochrane.org/handbook-diagnostic-test-accuracy/current. Accessed December 9, 2023.
[31] Morgano GP, Mbuagbaw L, Santesso N, Xie F, Brozek JL, Siebert U, et al. Defining decision thresholds for judgments on health benefits and harms using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) Evidence to Decision (EtD) frameworks: a protocol for a randomised methodological study (GRADE-THRESHOLD). BMJ Open 2022;12:e053246.
[32] Santesso N, Mustafa RA, Schünemann HJ, Arbyn M, Blumenthal PD, Cain J, et al. World Health Organization Guidelines for treatment of cervical intraepithelial neoplasia 2-3 and screen-and-treat strategies to prevent cervical cancer. Int J Gynaecol Obstet 2016;132:252−8.
[33] Samson D, Schoelles KM. Chapter 2: medical tests guidance (2) developing the topic and structuring systematic reviews of medical tests: utility of PICOTS, analytic frameworks, decision trees, and other frameworks. J Gen Intern Med 2012;27:11−9.
[34] Hsu J, Brożek JL, Terracciano L, Kreis J, Compalati E, Stein AT, et al. Application of GRADE: making evidence-based recommendations about diagnostic tests in clinical practice guidelines. Implement Sci 2011;6:62.
[35] Lord SJ, St John A, Bossuyt PM, Sandberg S, Monaghan PJ, O'Kane M, et al. Setting clinical performance specifications to develop and evaluate biomarkers for clinical use. Ann Clin Biochem 2019;56:527−35.
[36] Korevaar DA, Gopalakrishna G, Cohen JF, Bossuyt PM. Targeted test evaluation: a framework for designing diagnostic accuracy studies with clear study hypotheses. Diagn Progn Res 2019;3:22.
[37] Chu H, Cole SR. Sample size calculation using exact methods in diagnostic test studies. J Clin Epidemiol 2007;60:1201−2. author reply 2.
[38] Leeflang MM, Rutjes AW, Reitsma JB, Hooft L, Bossuyt PM. Variation of a test's sensitivity and specificity with disease prevalence. CMAJ 2013;185:E537−44.
[39] Chu H, Nie L, Cole SR, Poole C. Meta-analysis of diagnostic accuracy studies accounting for disease prevalence: alternative parameterizations and model selection. Stat Med 2009;28:2384−99.
[40] Leeflang MMSK, Scholten RJ, Davenport C. Cochrane handbook for systematic reviews of diagnostic test accuracy version 2. London: Cochrane; 2022: [Chapter 13]: Drawing conclusions.
[41] Yang B, Olsen M, Vali Y, Langendam MW, Takwoingi Y, Hyde CJ, et al. Study designs for comparative diagnostic test accuracy: a methodological review and classification scheme. J Clin Epidemiol 2021; 138:128−38.